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Lab 1: Introduction to SPSS

Introduction

The SPSS (Statistical Package for the Social Sciences) is a software for editing and analyzing all sorts of data. These data may come from basically any source: scientific research a customer database, etc. SPSS can open all file formats that are commonly used for structured data such as

1. i) Spreadsheets from Ms. Excel for open office
2. ii) Plain text files (.txt or .csv)
3. iii) Relational (SQL) database
4. iv) Stata and SAS

When you use SPSS, you work in one of several windows: the data view, the variable view, the output view, the draft output view and the script view. Eventually you will also use the syntax editor to save or refine your queries.

The data view: The data view displays your actual data and any new variables you have created.

The variable view: At the bottom of the data window, you will notice a table labeled variable view. The variable view window contains the definitions of each variable in your data set including its name, label, size, alignment, and other information.

SPSS output window view: The output window is where you see the result of your various queries such as frequency distributions, cross-tabs, statistical tests, and charts. If you have worked with excel, you are probably used to seeing all your work on one page, charts, data and calculations. In SPSS each window handles a separate task. The output window is where you see your result.

Advantages of using SPSS

1. Save time and effort, perform a job in a second that would require hours or days.
2. More exact calculations, avoiding rounding and approximations of manual calculation.
3. It allows working with large amount of data, using larger samples and including more variables.
4. It allows transferring the attention from the mechanical task of calculation to the conceptual tasks: decisions on the process, interpretation of the result and critical analysis.
5. SPSS is one of the most widely used tools in business research and other fields of social sciences.

It is compatible with many other data management programs. Therefore, it allows transmission of data from one program to another

Lab 2: Mean, Median, Mode, Standard Deviation, Percentile

Enter the following values in SPSS and Mean, Median, Mode, Standard Deviation and Percentiles:

Weight:	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	4	7	6	8	5	9

Output:

→ Frequencies

[DataSet1]

Statistics

midvalue

N	Valid	39
	Missing	0
Mean		32.69
Median		35.00
Mode		55
Std. Deviation		16.930
Percentiles	75	45.00

midvalue

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	5	4	10.3	10.3	10.3
	15	7	17.9	17.9	28.2
	25	6	15.4	15.4	43.6
	35	8	20.5	20.5	64.1
	45	5	12.8	12.8	76.9
	55	9	23.1	23.1	100.0
	Total	39	100.0	100.0	

Lab 3: Mean, Standard Deviation, Range

Enter the following values in SPSS and Calculate Mean, Standard Deviation, Range.

Weight: 10, 20, 30, 40, 50, 60

Output:

➔ **Frequencies**

Statistics

weight

N	Valid	6
	Missing	0
Mean		35.00
Std. Deviation		18.708
Range		50

weight

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	10	1	16.7	16.7	16.7
	20	1	16.7	16.7	33.3
	30	1	16.7	16.7	50.0
	40	1	16.7	16.7	66.7
	50	1	16.7	16.7	83.3
	60	1	16.7	16.7	100.0
	Total	6	100.0	100.0	

Lab 4: Correlation between X and Y

Enter the following values in SPSS and find the Correlation between X and Y

X	1	2	3	4	5	6	7
Y	4	6	1	2	3	5	7

Output:

Correlations

Correlations

		x	y
x	Pearson Correlation	1	.321
	Sig. (2-tailed)		.482
	N	7	7
y	Pearson Correlation	.321	1
	Sig. (2-tailed)	.482	
	N	7	7

Lab 5: Regression equation of Y on X

Enter the following values in SPSS and find the regression equation of y on x.

X	1	2	3	4	5	6	7
Y	4	6	5	2	1	2	3

Output:

→ Regression

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	x ^b		Enter

a. Dependent Variable: y

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.643 ^a	.414	.296	1.509

a. Predictors: (Constant), x

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	8.036	1	8.036	3.527	.119 ^b
	Residual	11.393	5	2.279		
	Total	19.429	6			

a. Dependent Variable: y

b. Predictors: (Constant), x

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	5.429	1.276		4.255	.008
	x	-.536	.285	-.643	-1.878	.119

a. Dependent Variable: y

Lab 6: Regression equation of X on Y

Enter the following values in SPSS and find the regression equation of x on y.

X	10	20	30	40	50	60	70
Y	20	70	40	10	60	50	30

Output:

→ Regression

[DataSet0]

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	y ^b	.	Enter

a. Dependent Variable: x

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.036 ^a	.001	-.198	23.649

a. Predictors: (Constant), y

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	3.571	1	3.571	.006	.939 ^b
	Residual	2796.429	5	559.286		
	Total	2800.000	6			

a. Dependent Variable: x

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	38.571	19.987		1.930	.112
	y	.036	.447	.036	.080	.939

a. Dependent Variable: x

Lab 7: Spearman's Rank correlation co-efficient (Non-repeated observation)

Calculate Spearman's Rank Correlation Coefficient from following data:

X	5	1	2	7	4	6	3
Y	2	1	3	4	5	7	6

(Rank for non-repeated observation)

Output:

RANK

Created Variables^a

Source Variable	Function	New Variable	Label
x ^b	Rank	Rx	Rank of x
y ^b	Rank	Ry	Rank of y

a. Mean rank of tied values is used for ties.

b. Ranks are in descending order.

NONPAR CORR

```
/VARIABLES=x y
/PRINT=SPEARMAN TWOTAIL NOSIG
/MISSING=PAIRWISE.
```

➔ Nonparametric Correlations

Correlations

			x	y
Spearman's rho	x	Correlation Coefficient	1.000	.464
		Sig. (2-tailed)	.	.294
		N	7	7
	y	Correlation Coefficient	.464	1.000
		Sig. (2-tailed)	.294	.
		N	7	7

Calculate Spearman's Rank Co-Relation Coefficient from following data:

Marks of A	55	24	42	43	45	31	20	39	21	66	64
Marks of B	75	23	34	40	54	33	21	30	36	27	14

(Rank for non-repeated observation)

Output:

RANK

Created Variables^a

Source Variable	Function	New Variable	Label
a ^b	Rank	Ra	Rank of a
b ^b	Rank	Rb	Rank of b

a. Mean rank of tied values is used for ties.
b. Ranks are in descending order.

Nonparametric Correlations

Correlations

			a	b
Spearman's rho	a	Correlation Coefficient	1.000	.164
		Sig. (2-tailed)	.	.631
		N	11	11
	b	Correlation Coefficient	.164	1.000
		Sig. (2-tailed)	.631	.
		N	11	11

Calculate Spearman's Rank Correlation Coefficient from following data:

Marks of X	106	143	140	150	154	114	122	134	142	166	146
Marks of Y	157	123	132	140	145	133	121	130	136	127	114

(Rank for non-repeated observation)

Output:

RANK

Created Variables^a

Source Variable	Function	New Variable	Label
x ^b	Rank	Rx	Rank of x
y ^b	Rank	Ry	Rank of y

a. Mean rank of tied values is used for ties.
b. Ranks are in descending order.

Nonparametric Correlations

Correlations

			x	y
Spearman's rho	x	Correlation Coefficient	1.000	-.118
		Sig. (2-tailed)	.	.729
		N	11	11
	y	Correlation Coefficient	-.118	1.000
		Sig. (2-tailed)	.729	.
		N	11	11

Lab 8: Spearman's Rank correlation co-efficient (Repeated observation)

Calculate Spearman's Rank Correlation Coefficient from following data:

Marks of X	186	164	157	104	164	180	157	104	155	164
Marks of Y	162	158	162	145	158	160	168	148	150	160

(Rank for repeated observation)

Output:

RANK

Created Variables^a

Source Variable	Function	New Variable	Label
x^b	Rank	Rx	Rank of x
y^b	Rank	Ry	Rank of y

a. Mean rank of tied values is used for ties.

b. Ranks are in descending order.

Nonparametric Correlations

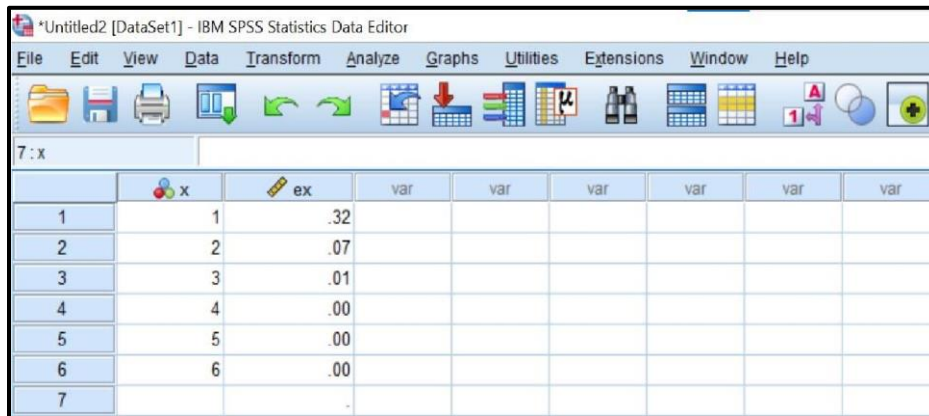
Correlations

			x	y
Spearman's rho	x	Correlation Coefficient	1.000	.576
		Sig. (2-tailed)	.	.081
		N	10	10
	y	Correlation Coefficient	.576	1.000
		Sig. (2-tailed)	.081	.
		N	10	10

Lab 9: Binomial Distribution

Using SPSS, Find the binomial distribution of a random variable x with $n = 10$ And $p = 0.05$, find

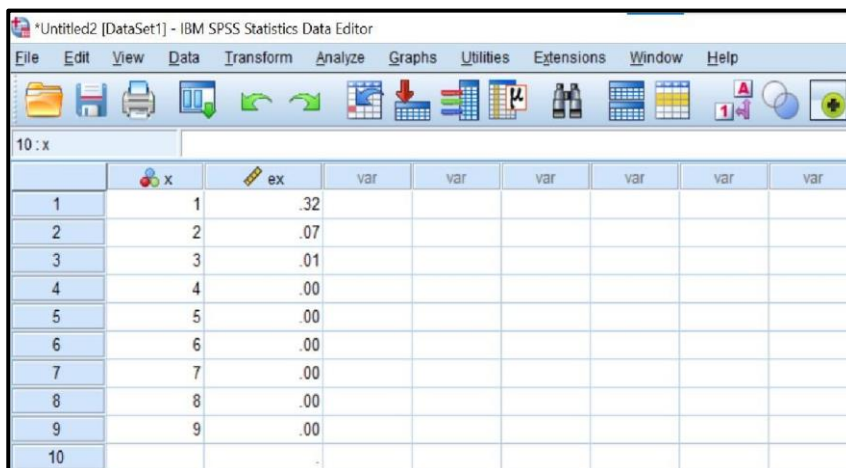
i) $p(X=6)$



7 : x								
	x	ex	var	var	var	var	var	var
1	1	.32						
2	2	.07						
3	3	.01						
4	4	.00						
5	5	.00						
6	6	.00						
7								

```
COMPUTE ex=PDF.BINOM(x,10,0.05).  
EXECUTE.
```

ii) $p(X=9)$



10 : x								
	x	ex	var	var	var	var	var	var
1	1	.32						
2	2	.07						
3	3	.01						
4	4	.00						
5	5	.00						
6	6	.00						
7	7	.00						
8	8	.00						
9	9	.00						
10								

```
COMPUTE ex=PDF.BINOM(x,10,0.05).  
EXECUTE.
```

iii) $p(X \leq 4)$

*Untitled2 [DataSet1] - IBM SPSS Statistics Data Editor

	x	ex	var	var	var	var	var	var
1	2	.07						
2	4	.00						
3	6	.00						
4	8	.00						
5	.	.						
6	.	.						

```
COMPUTE ex=PDF.BINOM(x,10,0.05)
EXECUTE.
```

iv) $p(X < 7)$

*Untitled2 [DataSet1] - IBM SPSS Statistics Data Editor

	x	ex	var	var	var	var	var	var
1	2	.07						
2	4	.00						
3	6	.00						
4	8	.00						
5	10	.00						
6	12	.00						
7	.	.						
8	.	.						

```
COMPUTE ex=PDF.BINOM(x,10,0.05)
EXECUTE.
```

Lab 10: Use of SPSS in RBD

Calculate RBD using SPSS from the following data:

A	52.50	E	46.30	D	44.10	C	48.10	B	40.90
D	44.20	B	42.90	A	51.30	E	49.30	C	32.60
B	49.10	A	47.30	C	38.10	D	41	E	47.20
C	43.20	D	42.50	E	67.20	B	55.10	A	45.30
E	47	C	43.2	B	46.7	A	46	D	43.2

Input:

LSD Input.sav [DataSet1] - IBM SPSS Statistics Data Editor

	Rows	Column	Treatment	Values
1	1	1	A	52.50
2	1	2	E	46.30
3	1	3	D	44.10
4	1	4	C	48.10
5	1	5	B	40.90
6	2	1	D	44.20
7	2	2	B	42.90
8	2	3	A	51.30
9	2	4	E	49.30
10	2	5	C	32.60
11	3	1	B	49.10
12	3	2	A	47.30
13	3	3	C	38.10
14	3	4	D	41.00
15	3	5	E	47.20
16	4	1	C	43.20
17	4	2	D	42.50
18	4	3	E	67.20
19	4	4	B	55.10
20	4	5	A	45.30
21	5	1	E	47.00
22	5	2	C	43.20
23	5	3	B	46.70
24	5	4	A	46.00
25	5	5	D	43.20

Output:

→ Univariate Analysis of Variance

[DataSet0]

Between-Subjects Factors

		N
Rows	1	5
	2	5
	3	5
	4	5
	5	5
Column	1	5
	2	5
	3	5
	4	5
	5	5
Treatment	A	5
	B	5
	C	5
	D	5
	E	5

Tests of Between-Subjects Effects

Dependent Variable: Values

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	673.075 ^a	12	56.090	2.213	.092
Intercept	53296.340	1	53296.340	2103.144	.000
Rows	141.078	4	35.270	1.392	.295
Column	183.758	4	45.940	1.813	.191
Treatment	348.238	4	87.060	3.435	.043
Error	304.095	12	25.341		
Total	54273.510	25			
Corrected Total	977.170	24			

a. R Squared = .689 (Adjusted R Squared = .378)

Post Hoc Tests

Rows

Multiple Comparisons

Dependent Variable: Values

LSD

(I) Rows	(J) Rows	Mean Difference (I- J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
1	2	2.3200	3.18379	.480	-4.6169	9.2569
	3	1.8400	3.18379	.574	-5.0969	8.7769
	4	-4.2800	3.18379	.204	-11.2169	2.6569
	5	1.1600	3.18379	.722	-5.7769	8.0969
2	1	-2.3200	3.18379	.480	-9.2569	4.6169
	3	-.4800	3.18379	.883	-7.4169	6.4569
	4	-6.6000	3.18379	.060	-13.5369	.3369
	5	-1.1600	3.18379	.722	-8.0969	5.7769
3	1	-1.8400	3.18379	.574	-8.7769	5.0969
	2	.4800	3.18379	.883	-6.4569	7.4169
	4	-6.1200	3.18379	.079	-13.0569	.8169
	5	-.6800	3.18379	.834	-7.6169	6.2569
4	1	4.2800	3.18379	.204	-2.6569	11.2169
	2	6.6000	3.18379	.060	-.3369	13.5369
	3	6.1200	3.18379	.079	-.8169	13.0569
	5	5.4400	3.18379	.113	-1.4969	12.3769
5	1	-1.1600	3.18379	.722	-8.0969	5.7769
	2	1.1600	3.18379	.722	-5.7769	8.0969
	3	.6800	3.18379	.834	-6.2569	7.6169
	4	-5.4400	3.18379	.113	-12.3769	1.4969

Based on observed means.

The error term is Mean Square(Error) = 25.341.

Homogeneous Subsets

Column

Multiple Comparisons

Dependent Variable: Values

LSD

(I) Column	(J) Column	Mean Difference (I- J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
1	2	2.7600	3.18379	.403	-4.1769	9.6969
	3	-2.2800	3.18379	.488	-9.2169	4.6569
	4	-.7000	3.18379	.830	-7.6369	6.2369
	5	5.3600	3.18379	.118	-1.5769	12.2969
2	1	-2.7600	3.18379	.403	-9.6969	4.1769
	3	-5.0400	3.18379	.139	-11.9769	1.8969
	4	-3.4600	3.18379	.298	-10.3969	3.4769
	5	2.6000	3.18379	.430	-4.3369	9.5369
3	1	2.2800	3.18379	.488	-4.6569	9.2169
	2	5.0400	3.18379	.139	-1.8969	11.9769
	4	1.5800	3.18379	.629	-5.3569	8.5169
	5	7.6400*	3.18379	.034	.7031	14.5769
4	1	.7000	3.18379	.830	-6.2369	7.6369
	2	3.4600	3.18379	.298	-3.4769	10.3969
	3	-1.5800	3.18379	.629	-8.5169	5.3569
	5	6.0600	3.18379	.081	-.8769	12.9969
5	1	-5.3600	3.18379	.118	-12.2969	1.5769
	2	-2.6000	3.18379	.430	-9.5369	4.3369
	3	-7.6400*	3.18379	.034	-14.5769	-.7031
	4	-6.0600	3.18379	.081	-12.9969	-.8769

Based on observed means.

The error term is Mean Square(Error) = 25.341.

*. The mean difference is significant at the 0.05 level.

Homogeneous Subsets

Treatment

Multiple Comparisons

Dependent Variable: Values

LSD

(I) Treatment	(J) Treatment	Mean Difference (I- J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
A	B	1.5400	3.18379	.637	-5.3969	8.4769
	C	7.4400*	3.18379	.038	.5031	14.3769
	D	5.4800	3.18379	.111	-1.4569	12.4169
	E	-2.9200	3.18379	.377	-9.8569	4.0169
B	A	-1.5400	3.18379	.637	-8.4769	5.3969
	C	5.9000	3.18379	.089	-1.0369	12.8369
	D	3.9400	3.18379	.240	-2.9969	10.8769
	E	-4.4600	3.18379	.187	-11.3969	2.4769
C	A	-7.4400*	3.18379	.038	-14.3769	-.5031
	B	-5.9000	3.18379	.089	-12.8369	1.0369
	D	-1.9600	3.18379	.550	-8.8969	4.9769
	E	-10.3600*	3.18379	.007	-17.2969	-3.4231
D	A	-5.4800	3.18379	.111	-12.4169	1.4569
	B	-3.9400	3.18379	.240	-10.8769	2.9969
	C	1.9600	3.18379	.550	-4.9769	8.8969
	E	-8.4000*	3.18379	.022	-15.3369	-1.4631
E	A	2.9200	3.18379	.377	-4.0169	9.8569
	B	4.4600	3.18379	.187	-2.4769	11.3969
	C	10.3600*	3.18379	.007	3.4231	17.2969
	D	8.4000*	3.18379	.022	1.4631	15.3369

Based on observed means.

The error term is Mean Square(Error) = 25.341.

*. The mean difference is significant at the 0.05 level.

Lab 11: Use of SPSS in CRD

The weight of a number of copper wires each of length 1m were obtained. These are shown below classified according to the die from which they come.

Die No.				
I	II	III	IV	V
2	0	4	3	2
4	7	1	1	4
4	5	3	5	2
7	6	0	3	5
4		5	4	
9		2		

Are the weights of copper wire for the different dies equal? Using SPSS:-

Input:

The screenshot displays the IBM SPSS Statistics Data Editor window. The title bar reads '*Untitled2 [DataSet1] - IBM SPSS Statistics Data Editor'. The menu bar includes File, Edit, View, Data, Transform, Analyze, Graphs, Utilities, Extensions, Window, and Help. The toolbar contains icons for file operations (e.g., Open, Save, Print), data manipulation (e.g., Paste, Copy), and visualization (e.g., Chart, Table). The data grid shows 27 rows and 8 columns. The first column is labeled 'Weight' and the second is 'dice'. The data is as follows:

	Weight	dice	var	var	var	var	var	var
1	2.0	1.00						
2	4.0	1.00						
3	8.0	1.00						
4	7.0	1.00						
5	4.0	1.00						
6	9.0	1.00						
7	.0	2.00						
8	7.0	2.00						
9	5.0	2.00						
10	6.0	2.00						
11	4.0	3.00						
12	1.0	3.00						
13	3.0	3.00						
14	.0	3.00						
15	5.0	3.00						
16	2.0	3.00						
17	3.0	4.00						
18	1.0	4.00						
19	5.0	4.00						
20	3.0	4.00						
21	4.0	4.00						
22	2.0	5.00						
23	4.0	5.00						
24	2.0	5.00						
25	5.0	5.00						
26								
27								

The status bar at the bottom shows 'Data View' and 'Variable View' tabs, with 'Data View' currently selected.

Output:

Output6.spv [Document2] - IBM SPSS Statistics Viewer

File Edit View Data Transform Insert Format Analyze Graphs Utilities Extensions Window Help

Output
Log
Oneway
Title
Notes
ANOVA

ONEWAY Weight BY VAR00003
/MISSING ANALYSIS.

➔ **Oneway**

ANOVA

Weight

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	35.977	4	8.994	1.810	.167
Within Groups	99.383	20	4.969		
Total	135.360	24			